# Transient Two-Dimensional Heat Conduction in a Rectangular Bar: Explicit FTCS and ADI Solutions

Samuel R. Denis*
*Department of Aerospace Engineering, Wichita State University, Wichita, Kansas, 67260*

**The transient temperature field in a long copper bar with fixed wall temperatures is computed with the explicit forward-time, centered-space (FTCS) scheme and the alternating direction implicit (ADI) scheme. The steady state is compared with the analytical solution on a grid of $31 \times 41$. With a time step of $0.2\,\mathrm{s}$ both schemes reach steady-state, which is defined as the total variation in temperature per step being less than $0.01\,^\circ\mathrm{C}$ defined as CONSS after about $410\,\mathrm{s}$ of simulated time. For both methods the max error is $0.2874\,^\circ\mathrm{C}$ near the hot corners where the wall discontinuity exists, root-mean-square error of $0.0260\,^\circ\mathrm{C}$, mean error of $-0.0011\,^\circ\mathrm{C}$. With a time step of $1.0\,\mathrm{s}$, FTCS diverges, as its stability limit predicts, ADI converges in 496 steps. The steady-state wall heat flux satisfies the overall energy balance of the system within $0.1\,\%$.**

## Nomenclature

| | | |
|---|---|---|
| $L$, $W$ | = | bar length in $x$ and width in $y$, m |
| $x$, $y$ | = | coordinates, m |
| $IM$, $JM$ | = | number of grid points in $x$ and $y$ |
| $T$ | = | temperature, °C |
| $T_0$ | = | initial interior temperature, °C |
| $T_1$, $T_2$, $T_3$, $T_4$ | = | bottom, left, top, and right wall temperatures, °C |
| $T_A$, $T_B$ | = | parts of the analytical solution driven by $T_1$ and $T_3$, °C |
| $\alpha$ | = | thermal diffusivity, $\mathrm{m^2\,s^{-1}}$ |
| $k$ | = | thermal conductivity, $\mathrm{W\,m^{-1}\,^\circ C^{-1}}$ |
| $t$ | = | time, s |
| $\Delta t$ | = | time step, s |
| $\Delta x$, $\Delta y$ | = | grid spacing in $x$ and $y$, m |
| $i$, $j$ | = | grid indices in $x$ and $y$ |
| $n$ | = | time level |
| $d_x$, $d_y$ | = | FTCS diffusion numbers, $\alpha\Delta t/\Delta x^2$ and $\alpha\Delta t/\Delta y^2$ |
| $d_1$, $d_2$ | = | ADI half-step coefficients, $\alpha\Delta t/(2\Delta x^2)$ and $\alpha\Delta t/(2\Delta y^2)$ |
| $G$ | = | amplification factor |
| $\theta$, $\phi$ | = | phase angles of a Fourier mode in $x$ and $y$ |
| TV | = | total variation, sum of $\lvert T_{i,j}^{n+1} - T_{i,j}^{n}\rvert$ over interior nodes, °C |
| CONSS | = | steady-state tolerance on TV, °C |
| $m$ | = | series term index |
| $\hat{\mathbf{n}}_{\mathrm{in}}$ | = | inward unit normal of a wall |
| $q_{\mathrm{in}}$ | = | heat flux into the bar, $\mathrm{W\,m^{-2}}$ |

## I. Introduction

This report presents solutions to Problems 3.5 through 3.8 of Hoffmann and Chiang [1]. Problem 3.5 describes transient heat conduction in a long rectangular copper bar, initially at a uniform temperature, whose four sides are held at constant temperatures. The transient solution is computed with the explicit forward-time, centered-space (FTCS)

*Graduate Student, Department of Aerospace Engineering, Wichita State University.

scheme, the alternating direction implicit (ADI) scheme, or the fractional step method (FSM), and the steady state is compared with the analytical series solution. The steady-state heat flux is evaluated along the sides $y = 0$, $y = W$, and $x = 0$. The ADI method was selected over FSM. Finally, the accuracy, stability, and efficiency of the two schemes are compared.

## II. Problem Description

The geometry shown in Fig. 1 defines the axis system as $x$-right and $y$-up with the length and width of the bar defined as $L$ and $W$, respectively. The figure defines the indices $i$ and $j$ in base one whereas in the solution provided all indices are given in base 0 notation with the ends of the bar defined as $IM$ and $JM$, respectively.

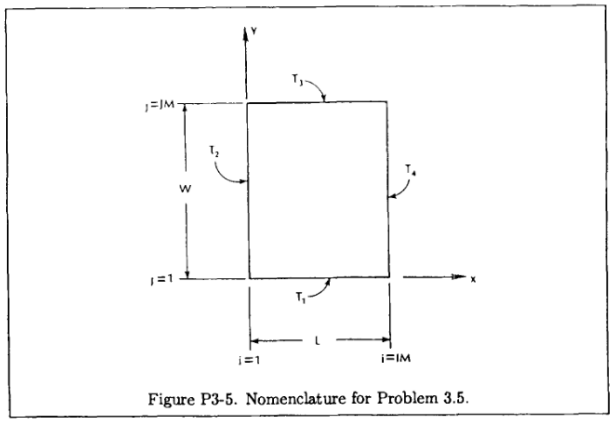


Figure P3-5. Nomenclature for Problem 3.5.

**Fig. 1 Geometry, axes, and boundary conditions of the bar (after Ref. [1]).**

The bar is copper, with the parameters

$$\begin{gathered} L = 0.3\,\text{m}, \quad W = 0.4\,\text{m}, \quad \alpha = 11.234 \times 10^{-5}\,\text{m}^2\,\text{s}^{-1}, \quad k = 380\,\text{W}\,\text{m}^{-1}\,^\circ\text{C}^{-1} \\ T_0 = 0\,^\circ\text{C}, \quad T_1 = 40\,^\circ\text{C}, \quad T_2 = 0\,^\circ\text{C}, \quad T_3 = 10\,^\circ\text{C}, \quad T_4 = 0\,^\circ\text{C} \end{gathered} \tag{1}$$

The computational grid has uniform spacing, with base-0 indices:

$$x_i = i\,\Delta x, \quad i = 0, \ldots, IM-1; \qquad y_j = j\,\Delta y, \quad j = 0, \ldots, JM-1 \tag{2}$$

$$\Delta x = \frac{L}{IM-1} = 0.01\,\text{m}, \qquad \Delta y = \frac{W}{JM-1} = 0.01\,\text{m}, \qquad IM = 31,\ JM = 41 \tag{3}$$

It is assumed that the solution has reached steady state if the total variation in temperature from one time point to the next is less than CONSS. CONSS is given as 0.01 °C in the problem statement.

## III. Assumptions

- The thermal conductivity $k$ and diffusivity $\alpha$ are constant.
- The bar is long, so the problem is two-dimensional, with no conduction along its axis.
- There is no internal heat generation.
- The wall temperatures are fixed for $t > 0$.
- Each corner node is set to the average of the two adjacent wall temperatures; the stencil never reads these nodes.
- Steady state is defined as TV < CONSS, Eq. (25).
- All arithmetic is in double precision.

## IV. Governing Equations and Boundary Conditions

The temperature in the bar satisfies the two-dimensional heat conduction equation

$$\frac{\partial T}{\partial t} = \alpha\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right), \qquad 0 < x < L, \quad 0 < y < W, \quad t > 0 \tag{4}$$

with the initial condition

$$T(x, y, 0) = T_0 \tag{5}$$

and the boundary conditions

$$\begin{aligned} T(x, 0, t) &= T_1 \quad \text{(bottom)} \\ T(0, y, t) &= T_2 \quad \text{(left)} \\ T(x, W, t) &= T_3 \quad \text{(top)} \\ T(L, y, t) &= T_4 \quad \text{(right)} \end{aligned} \qquad t > 0 \tag{6}$$

## V. Numerical Methods

### A. Explicit FTCS Scheme

The FTCS method is derived from Eq. (4) by substituting a forward difference approximation for the time derivative and central difference approximations for the spatial derivatives, which gives Eq. (7).

$$T_{i,j}^{n+1} = T_{i,j}^{n} + d_x \left( T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n} \right) + d_y \left( T_{i,j+1}^{n} - 2T_{i,j}^{n} + T_{i,j-1}^{n} \right) \tag{7}$$

The diffusion numbers are given by Eq. (8); for this problem they are constants.

$$d_x = \frac{\alpha \, \Delta t}{(\Delta x)^2}, \qquad d_y = \frac{\alpha \, \Delta t}{(\Delta y)^2} \tag{8}$$

The time derivative is first-order accurate and the central differences are second-order accurate, so the scheme is accurate to $O[\Delta t, (\Delta x)^2, (\Delta y)^2]$. The problem statement gives $\Delta x = \Delta y = 0.01$ m.

The FTCS method is explicit, which means that Eq. (7) can be applied directly to each node: each new value depends only on known values at time level $n$, so no simultaneous system has to be solved. This property allows the algorithm to be trivially parallelized for large systems. The algorithm defines the constants and an initial array of nodes with the initial and boundary temperatures applied, then loops over the interior nodes, $j = 1, \ldots, JM - 2$ and $i = 1, \ldots, IM - 2$, solving Eq. (7) and summing TV. After each pass, the pointers to the current and next arrays are swapped, and Eq. (25) is evaluated to determine when the loop can terminate. If TV is not finite, the algorithm exits with a diverged status. A step limit, set arbitrarily at 10,000 steps, prevents infinite loops.

Applying the von Neumann stability analysis to Eq. (7), with a Fourier component of phase angles $\theta$ and $\phi$ in $x$ and $y$, gives the amplification factor [1, Sec. 4.4]

$$G = 1 + 2d_x(\cos\theta - 1) + 2d_y(\cos\phi - 1) \tag{9}$$

Stability requires $|G| \leq 1$ for all $\theta$ and $\phi$, which gives [1, Eq. (4-71)]

$$d_x + d_y \leq \frac{1}{2} \tag{10}$$

With $d_x = d_y = d$, the stability requirement becomes $d \leq 0.25$. Hoffmann and Chiang [1] note that this severe restriction on the step size can make the method inefficient for some applications. For this grid, the stability condition implies

$$\Delta t \leq \frac{(\Delta x)^2}{4\alpha} = 0.2225 \, \text{s} \tag{11}$$

so $\Delta t = 1.0$ s lies outside the stable range.

### B. Alternating Direction Implicit Scheme

Hoffmann and Chiang [1] describe how a two-dimensional implicit scheme, through its five-point stencil, produces a pentadiagonal system of equations. The pentadiagonal system is computationally expensive to solve, which led to the ADI method. The ADI method splits the pentadiagonal system into two tridiagonal systems by using a half step in time, which makes the $x$ and $y$ sweeps separable. The $x$ and $y$ sweeps are given by Eqs. (12) and (13), respectively.

$$-d_1 T_{i-1,j}^{*} + (1 + 2d_1) T_{i,j}^{*} - d_1 T_{i+1,j}^{*} = d_2 T_{i,j-1}^{n} + (1 - 2d_2) T_{i,j}^{n} + d_2 T_{i,j+1}^{n} \tag{12}$$

$$-d_2 T_{i,j-1}^{n+1} + (1 + 2d_2) T_{i,j}^{n+1} - d_2 T_{i,j+1}^{n+1} = d_1 T_{i-1,j}^{*} + (1 - 2d_1) T_{i,j}^{*} + d_1 T_{i+1,j}^{*} \tag{13}$$

The implicit scheme is accurate to $O[(\Delta t)^2, (\Delta x)^2, (\Delta y)^2]$ and is unconditionally stable [1]. This is advantageous compared with the FTCS scheme, as it is more accurate and stable.

The solution starts by defining the half-step coefficients, Eq. (14), which are constant because $\Delta t$, $\Delta x$, and $\Delta y$ are known.

$$d_1 = \frac{1}{2}\frac{\alpha\,\Delta t}{(\Delta x)^2} = \frac{d_x}{2}, \qquad d_2 = \frac{1}{2}\frac{\alpha\,\Delta t}{(\Delta y)^2} = \frac{d_y}{2} \tag{14}$$

As in the FTCS scheme, $d_x = d_y = d$. For $\Delta t = 0.2\,\text{s}$, $d = 0.2247$, and for $\Delta t = 1.0\,\text{s}$, $d = 1.1234$; $d_x + d_y$ is 0.449 and 2.247, respectively, and the ADI coefficients $d_1 = d_2$ are 0.1123 and 0.5617, respectively.

The initial conditions are applied to the current, next, and $T^*$ arrays of nodes, after which the $x$ sweep and then the $y$ sweep are solved. Both sweeps follow the same general process. The $x$ sweep is implicit in $x$ and explicit in $y$: each interior row, $j = 1, \ldots, JM - 2$, is set up as a system of equations following Eq. (12) and solved with the Thomas algorithm discussed in Sec. V.C. The intermediate half-step results are stored in the $T^*$ array. The $y$ sweep follows the same process, setting up the system for each column following Eq. (13) from $T^*$ and passing it to the Thomas algorithm. The solved nodes are then written to the next array. After both sweeps, the pointers to the current and next arrays are swapped, the step is incremented, and the same exit conditions as in the FTCS scheme are checked.

### C. Thomas Algorithm

The ADI method as discussed splits the pentadiagonal system into a pair of tridiagonal systems of equations. The naive approach to solving a system of equations would be to employ Gaussian elimination which has a big O complexity of $O(n^3)$. To efficiently solve the special case of a tridiagonal system Eq. (15) the Thomas Algorithm [2] was developed. Hoffmann and Chiang [1] demonstrate the same algorithm in Appendix B under a different name. This approach takes advantage of only three diagonals being nonzero to solve the system in $O(n)$ complexity. The notation below matches the code provided.

$$a_k\,x_{k-1} + b_k\,x_k + c_k\,x_{k+1} = d_k, \qquad k = 0, 1, \ldots, n-1, \qquad a_0 = c_{n-1} = 0 \tag{15}$$

$$\begin{bmatrix} b_0 & c_0 & & & \\ a_1 & b_1 & c_1 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{n-2} & b_{n-2} & c_{n-2} \\ & & & a_{n-1} & b_{n-1} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-2} \\ x_{n-1} \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ \vdots \\ d_{n-2} \\ d_{n-1} \end{bmatrix} \tag{16}$$

where the $a_k$, $b_k$, $c_k$ and $d_k$ are known, solving for the unknown $x_k$. For each time step the inner nodes of a row form a system of equations Eq. (16). The algorithm begins with a linear forwards sweep handling the first equation which does not have $a_k$ by using Eq. (17). Then the rest of the rows forward sweep is solved of the form Eq. (18).

$$s_0 = \frac{c_0}{b_0}, \qquad x'_0 = \frac{d_0}{b_0} \tag{17}$$

$$\left.\begin{aligned} \delta_k &= b_k - a_k\,s_{k-1} \\ s_k &= \frac{c_k}{\delta_k}, \qquad k \le n-2 \\ x'_k &= \frac{d_k - a_k\,x'_{k-1}}{\delta_k} \end{aligned}\right\} \qquad k = 1, 2, \ldots, n-1 \tag{18}$$

The loop completes over the inner nodes and then back substitutes following Eq. (19).

$$x_{n-1} = x'_{n-1}, \qquad x_k = x'_k - s_k\,x_{k+1}, \qquad k = n-2, n-3, \ldots, 0 \tag{19}$$

The algorithm returns the solution $x_k$ for the row.

### D. Analytical Steady-State Solution

At steady state, $\partial T/\partial t = 0$ and Eq. (4) reduces to Laplace's equation,

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \tag{20}$$

for which the problem statement gives the series solution

$$T(x, y) = T_A(x, y) + T_B(x, y) \tag{21}$$

$$T_A = 2\,T_1 \sum_{m=1}^{\infty} \frac{1 - \cos(m\pi)}{m\pi} \, \frac{\sinh[m\pi(W - y)/L]}{\sinh(m\pi W/L)} \, \sin\!\left(\frac{m\pi x}{L}\right) \tag{22}$$

$$T_B = 2\,T_3 \sum_{m=1}^{\infty} \frac{1 - \cos(m\pi)}{m\pi} \, \frac{\sinh(m\pi y/L)}{\sinh(m\pi W/L)} \, \sin\!\left(\frac{m\pi x}{L}\right) \tag{23}$$

The analytical steady state solution provided by the problem statement can be simplified for the case of $T_2 = T_4 = 0$ as Eq. (24). The solution below is continuous and was evaluated at the same grid of nodes as the solvers for consistency of comparison. Numerical stability of the hyperbolic sine had to be taken into account due to double precision floating point overflow. The identity $\sinh a/\sinh b = e^{a-b}(1 - e^{-2a})/(1 - e^{-2b})$, for $0 \le a \le b$, enables the stable calculation of Eq. (24). By choosing an odd $m$ for the series expansion the same result can be computed in half of the work. The factor $2\,(1 - \cos m\pi)$ becomes a constant 4. This works for the case of a mirror symmetric boundary condition, where the antisymmetric even terms are exactly zero. The series is truncated when the bound on the next term, its amplitude times the sinh ratio without the sine, falls below $1 \times 10^{-8}$ °C. The sine is left out of the bound so that a node where $\sin(m\pi x/L) = 0$ cannot stop the series early.

$$T(x, y) = \frac{4}{\pi} \sum_{m=1,3,5,\ldots} \frac{1}{m} \left[ T_1 \frac{\sinh[m\pi(W - y)/L]}{\sinh(m\pi W/L)} + T_3 \frac{\sinh(m\pi y/L)}{\sinh(m\pi W/L)} \right] \sin\!\left(\frac{m\pi x}{L}\right) \tag{24}$$

### E. Convergence and Wall Heat Flux

The steady state criterion Eq. (25) defined by Hoffmann and Chiang [1] in the problem statement defines the convergence across the nodes required for the time-marching FTCS and ADI to terminate.

$$\mathrm{TV} = \sum_{i=1}^{IM-2} \sum_{j=1}^{JM-2} \left| T_{i,j}^{n+1} - T_{i,j}^{n} \right| < \mathrm{CONSS}, \qquad \mathrm{CONSS} = 0.01\,^{\circ}\mathrm{C} \tag{25}$$

The wall heat flux follows from Fourier's law, taking the component along the inward normal of each wall so that a positive $q_{\mathrm{in}}$ is heat entering the bar:

$$\mathbf{q} = -k\,\nabla T, \qquad q_{\mathrm{in}} = \mathbf{q} \cdot \hat{\mathbf{n}}_{\mathrm{in}} \tag{26}$$

The wall gradients are evaluated with the second-order one-sided differences of Eq. (27), the forward form [1, Eq. (2-23)] at $y = 0$ and the backward form [1, Table 2.5] at $y = W$, both $O[(\Delta y)^2]$. The left wall, $x = 0$, uses the same forward form in $x$. The corners are excluded because the wall temperature jumps there.

$$\left.\frac{\partial T}{\partial y}\right|_{y=0} \approx \frac{-3T_{i,0} + 4T_{i,1} - T_{i,2}}{2\Delta y}, \qquad \left.\frac{\partial T}{\partial y}\right|_{y=W} \approx \frac{3T_{i,JM-1} - 4T_{i,JM-2} + T_{i,JM-3}}{2\Delta y} \tag{27}$$

## VI. Results

### A. Analytical Baseline

The baseline result shown in Fig. 2 was computed on the same grid of nodes as the FTCS and ADI solvers for ease of comparison. It defines the steady-state baseline that the solvers are compared against for correctness.

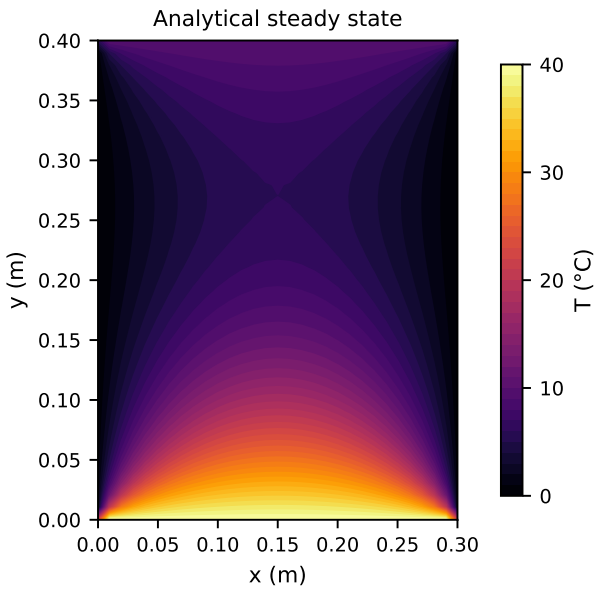



**Fig. 2 Analytical steady-state temperature field.**

## B. FTCS

*1. Transient Solution*

The FTCS solution is plotted in Fig. 3 at $t = 10\,\text{s}$, $40\,\text{s}$, and $410.8\,\text{s}$ (steady state) per the problem statement. It demonstrates the progression of heat from the boundary into the interior over time and is consistent with the expected result.

At $t = 10\,\text{s}$ and $40\,\text{s}$, Fig. 4 shows the heat diffusing into the interior of the cool bar. The values plotted at the corners are an artifact of the boundary conditions applied and do not represent the physics involved. The left-to-right symmetry of the problem statement can also be seen.

*2. Stability*

The FTCS method has been shown to be stable for this problem at $\Delta t = 0.2\,\text{s}$ converging in 2054 steps. For $\Delta t = 1.0\,\text{s}$, Fig. 5 shows it to be unstable: the temperature at the sensor location $(x, y) = (0.10, 0.05)$ m grows exponentially, demonstrating numerical instability in the solver. The run stopped at step 344 due to TV overflow. The probe temperature grew and the sign of the temperature flipped every step which is clearly physically false.

*3. Temperature Histories and Convergence*

Figure 6 demonstrates sensors embedded in the bar, which allows us to observe the change in temperature at individual locations over time. The probes reached 90 % of their steady value at 52, 97, and 142 s. The final values converged to within $0.021\,^\circ\text{C}$ of the exact solution. TV decayed at a rate of $0.019\,27\,\text{s}^{-1}$ as shown in Fig. 7.

*4. Steady State Compared with the Analytical Solution*

Evaluating the FTCS method against the analytical solution in Fig. 8, the maximum error for $\Delta t = 0.2\,\text{s}$ is $0.2874\,^\circ\text{C}$ at $(x, y) = (0.02, 0.01)$ m, near the hot corner. The RMS error is $0.0260\,^\circ\text{C}$, with a mean of $-0.0011\,^\circ\text{C}$. At the center node, the temperature is $7.683\,55\,^\circ\text{C}$ versus $7.683\,26\,^\circ\text{C}$, a difference of $0.0003\,^\circ\text{C}$. Figure 9 shows the error of FTCS versus the analytical solution along the centerline.

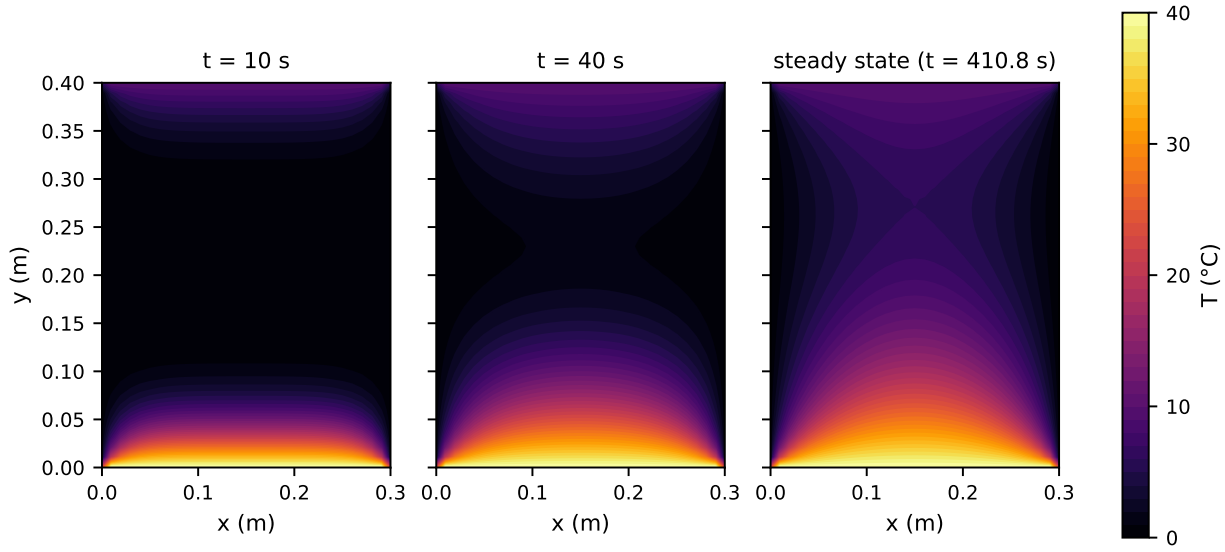



**Fig. 3 FTCS temperature field at $t = 10\,\mathrm{s}$, $t = 40\,\mathrm{s}$, and steady state, $\Delta t = 0.2\,\mathrm{s}$.**

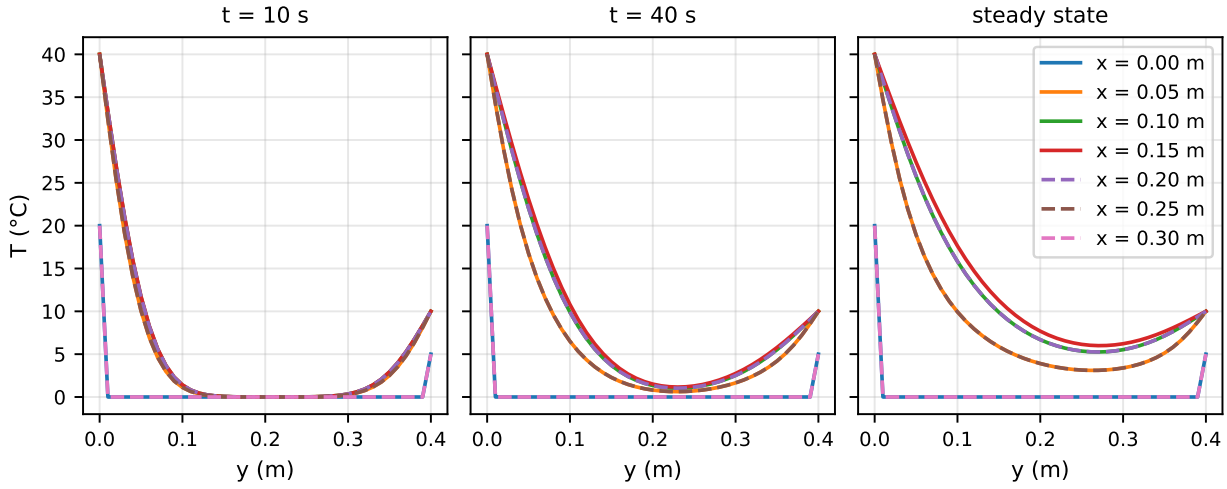



**Fig. 4 FTCS temperature versus $y$ at $x = 0$ to $0.30\,\mathrm{m}$ in steps of $0.05\,\mathrm{m}$, $\Delta t = 0.2\,\mathrm{s}$.**

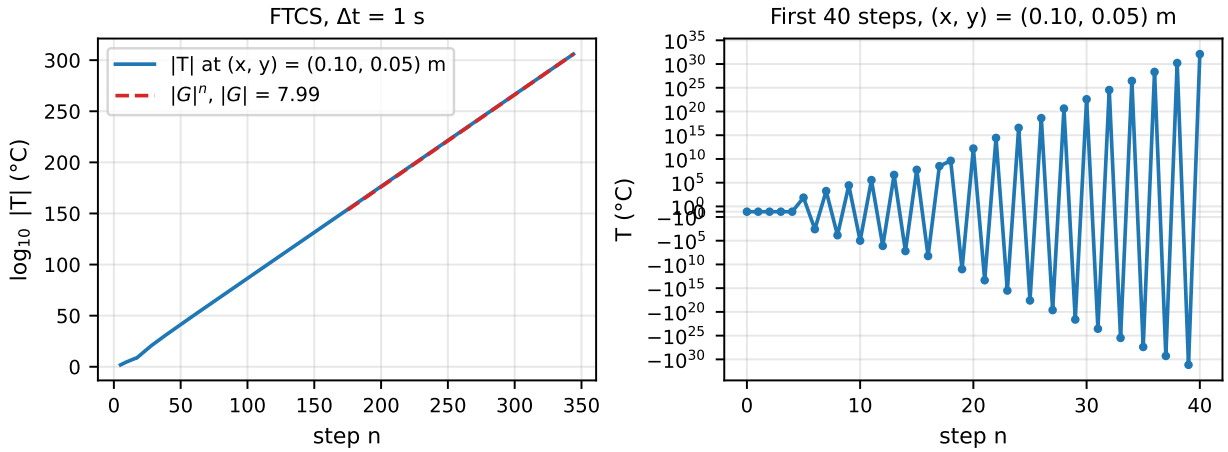



**Fig. 5 FTCS with $\Delta t = 1.0\,\mathrm{s}$: growth of the probe temperature compared with the amplification factor $|G|^n$, and the first 40 steps.**

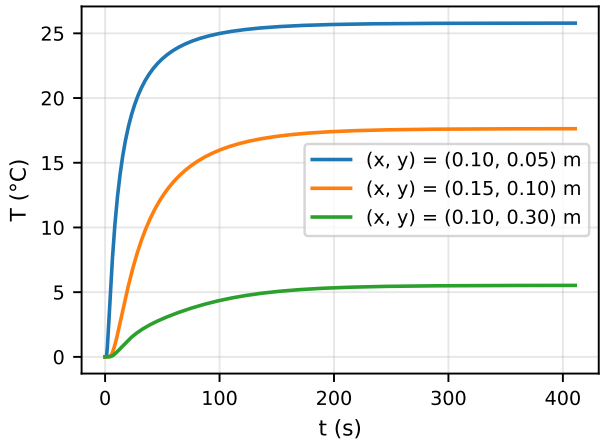



**Fig. 6 FTCS temperature histories at the three probe locations, $\Delta t = 0.2\,\mathrm{s}$.**

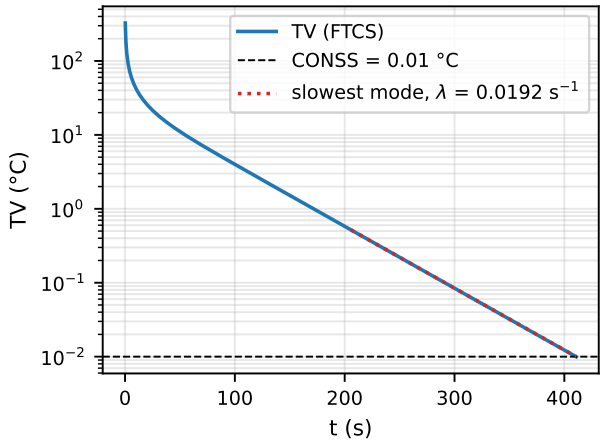



**Fig. 7 FTCS total variation versus time with the steady-state tolerance and the slowest-mode decay rate.**

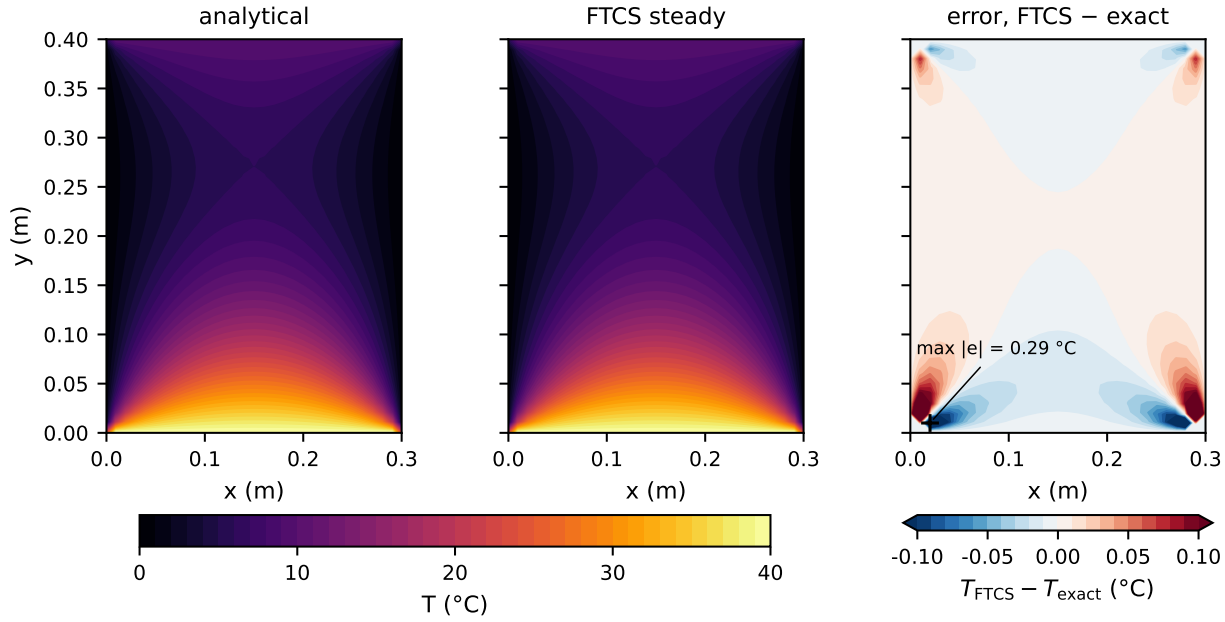



**Fig. 8 Analytical and FTCS steady-state fields and their difference.**

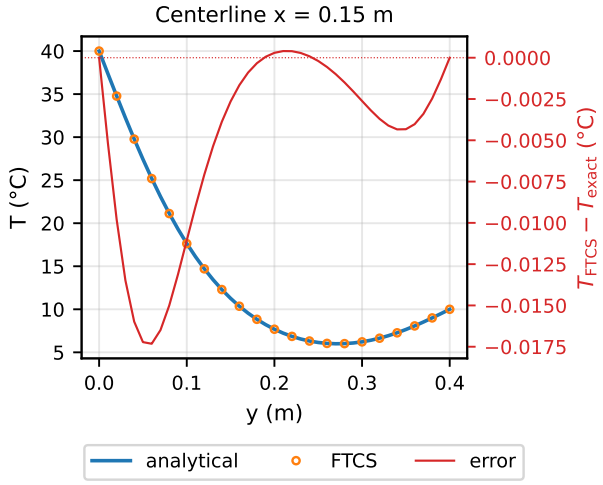



**Fig. 9 Steady-state temperature along the centerline $x = 0.15$ m, with the FTCS error.**

## C. ADI

*1. Transient Solution*

The ADI solution is plotted in Fig. 10 in the same fashion as FTCS and shows the same progression to steady state. The convergence criterion, Eq. (25), is met slightly later than for the FTCS method, after 2057 steps. The ADI profiles agree with the FTCS profiles shown in Fig. 4 to within 0.1 °C, so they are not repeated.

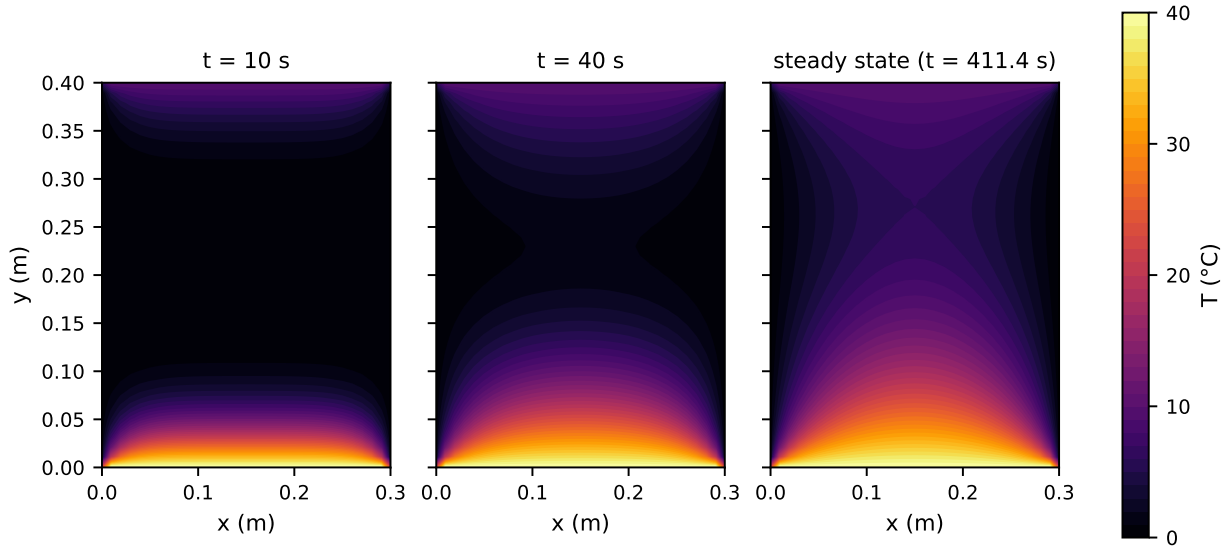



**Fig. 10 ADI temperature field at $t = 10\,\text{s}$, $t = 40\,\text{s}$, and steady state, $\Delta t = 0.2\,\text{s}$.**

*2. Temperature Histories and Convergence*

For $\Delta t = 1.0\,\text{s}$, ADI converged in 496 steps, whereas FTCS diverged. At that time step, ADI gave probe histories nearly identical to those at $\Delta t = 0.2\,\text{s}$ shown in Fig. 11. The convergence for $\Delta t = 0.2\,\text{s}$, shown in Fig. 12, is essentially the same as the FTCS result.

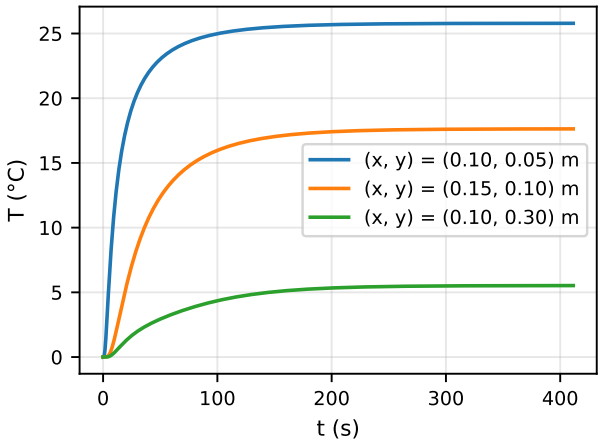



**Fig. 11 ADI temperature histories at the three probe locations, $\Delta t = 0.2\,\text{s}$.**

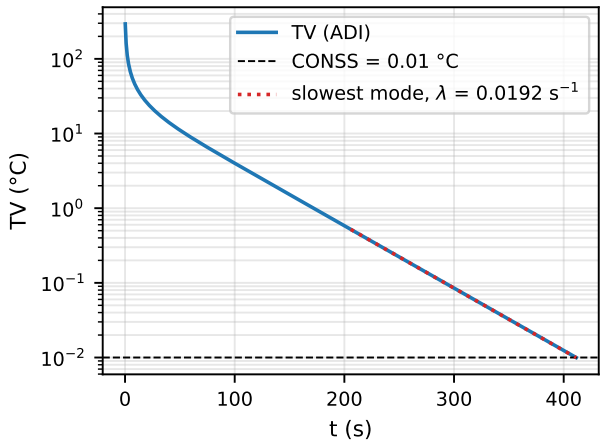



**Fig. 12 ADI total variation versus time with the steady-state tolerance and the slowest-mode decay rate.**

*3. Steady State Compared with the Analytical Solution*

Evaluating the ADI method against the analytical solution in Fig. 13, the maximum error for $\Delta t = 0.2\,\mathrm{s}$ is $0.2874\,^{\circ}\mathrm{C}$ at $(x, y) = (0.02, 0.01)$ m, similar to the FTCS result. The RMS error is $0.0260\,^{\circ}\mathrm{C}$, with a mean of $-0.0011\,^{\circ}\mathrm{C}$. At the center node, the temperature is $7.683\,53\,^{\circ}\mathrm{C}$ versus $7.683\,26\,^{\circ}\mathrm{C}$, a difference of $0.0003\,^{\circ}\mathrm{C}$.

For $\Delta t = 1.0\,\mathrm{s}$, ADI stopped at $t = 496\,\mathrm{s}$. The maximum error is $0.2873\,^{\circ}\mathrm{C}$ at the same node, the RMS error is $0.0260\,^{\circ}\mathrm{C}$, and the mean is $0.0007\,^{\circ}\mathrm{C}$. At the center node, the temperature is $7.6878\,^{\circ}\mathrm{C}$ versus $7.6833\,^{\circ}\mathrm{C}$, a difference of $0.0046\,^{\circ}\mathrm{C}$.

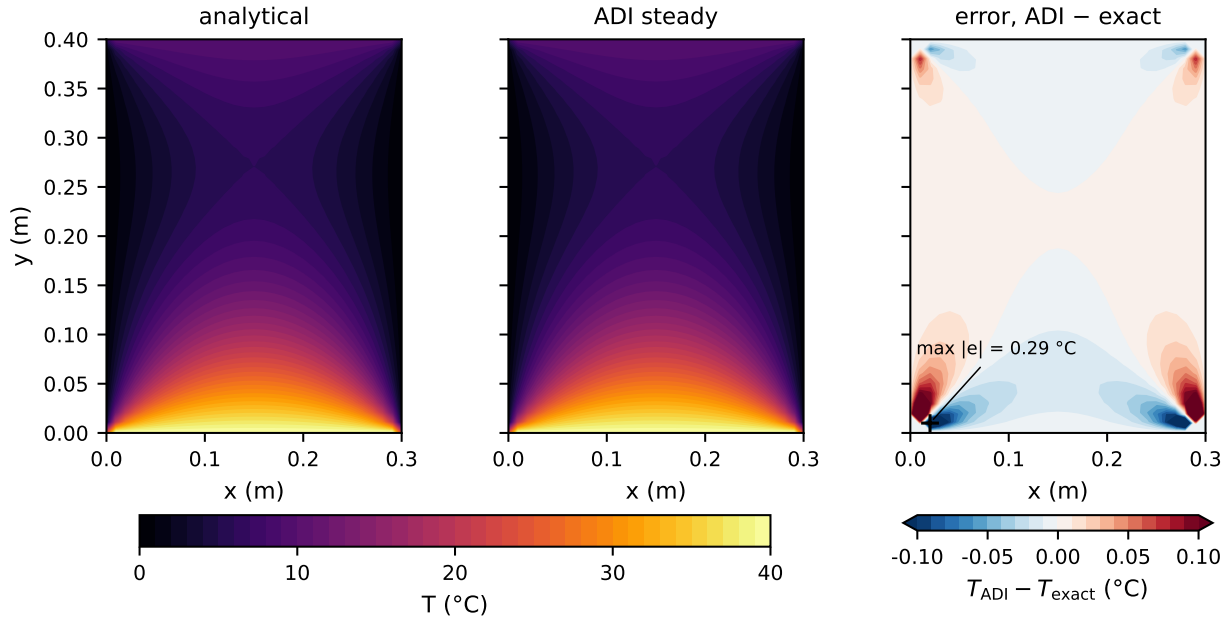



**Fig. 13 Analytical and ADI steady-state fields and their difference.**

## D. Wall Heat Flux

Evaluating the steady-state wall heat flux in Fig. 14, the heat flux into the bar at the middle of the bottom wall, $x = 0.15\,\mathrm{m}$, is $100.46\,\mathrm{kW\,m^{-2}}$ for FTCS versus $100.26\,\mathrm{kW\,m^{-2}}$ for the analytical field, a difference of $0.21\,\mathrm{kW\,m^{-2}}$.

At the middle of the top wall, the heat flux is 19.38 kW m$^{-2}$ versus 19.32 kW m$^{-2}$, and at the middle of the left wall, $y = 0.20$ m, it is $-31.37$ kW m$^{-2}$ versus $-31.34$ kW m$^{-2}$. Excluding the two nodes nearest each corner, the maximum difference from the analytical field is 12.05 kW m$^{-2}$ on the bottom wall, 3.01 kW m$^{-2}$ on the top wall, and 12.03 kW m$^{-2}$ on the left wall. The ADI heat flux agrees with FTCS to within 0.08 W m$^{-2}$.

Integrated along each wall, the FTCS heat flow per unit depth is 60.89 kW m$^{-1}$ into the bar through the bottom wall and 14.13 kW m$^{-1}$ through the top wall, and 37.53 kW m$^{-1}$ out through each of the left and right walls. The net is $-40.0$ W m$^{-1}$, or 0.07 % of the heat entering through the bottom wall. The analytical field gives a net of $-52.0$ W m$^{-1}$, or 0.09 %. The corners must be excluded from the analysis because the boundary conditions at those locations are not physically accurate: the wall temperature jumps there, so the flux is unbounded. The exact flux omits the same locations, and excluding them is why the balance is only approximate.

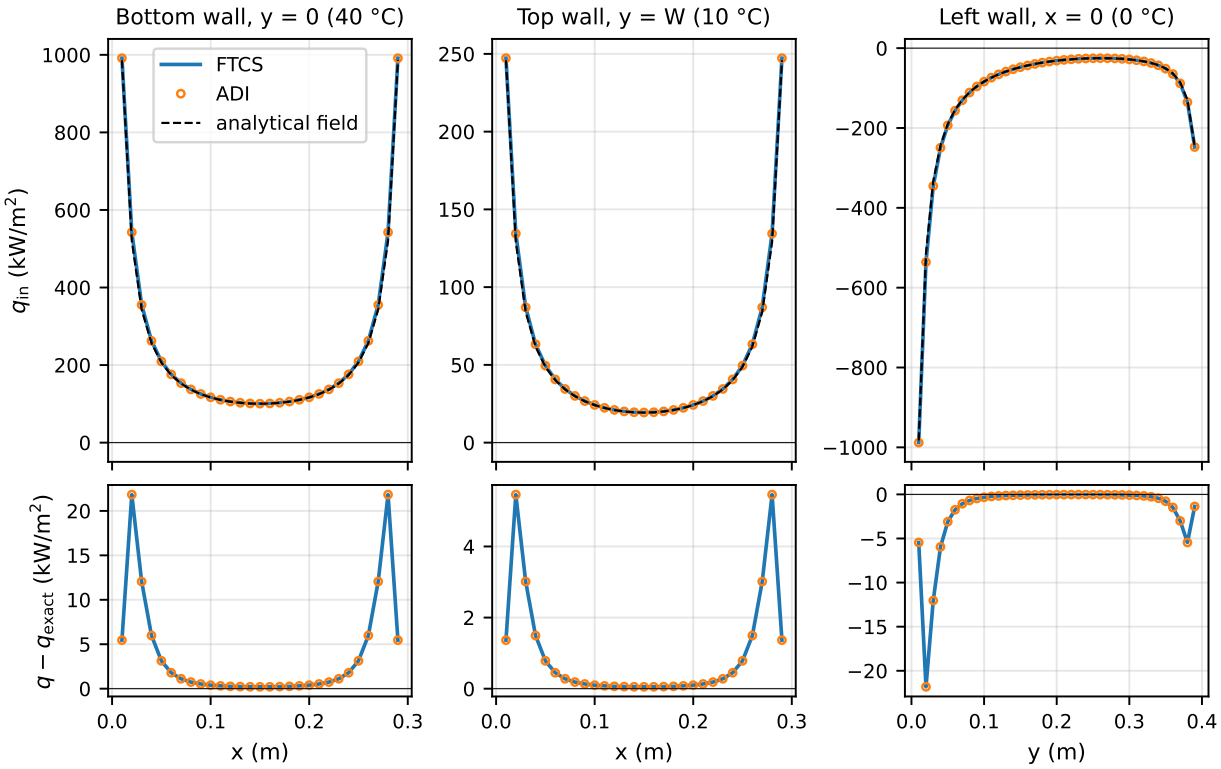



**Fig. 14 Steady-state heat flux into the bar along the bottom, top, and left walls (top row) and the difference from the analytical field (bottom row). Corners are excluded; the flux there is unbounded.**

## VII. Analysis

### A. Stability

The diffusion number is $d = 0.2247$ at $\Delta t = 0.2$ s and $d = 1.1234$ at $\Delta t = 1.0$ s. The worst case of Eq. (9) is $\theta = \phi = \pi$. With $\cos\theta = \cos\phi = -1$ and $d_x = d_y = d$, Eq. (9) becomes

$$G = 1 - 4d - 4d = 1 - 8d \tag{28}$$

At $\Delta t = 0.2$ s, $G = -0.797$: $|G| < 1$, so the scheme is stable, but the high-frequency error alternates in sign. At $\Delta t = 1.0$ s, $G = -7.99$: $|G| > 1$, so the scheme is unstable, consistent with the limit $\Delta t \le 0.2225$ s of Eq. (11). The measured growth of the probe temperature in Fig. 5 is 0.899 decades per step, against $\log_{10}|G| = 0.902$ predicted. The probe overshoot of 71.6 °C violates the maximum principle, which requires the temperature to stay between the coldest and hottest boundary values, 0 °C and 40 °C.

### B. Error

Both methods give the same steady-state error: an RMS error of 0.026 °C and a maximum of 0.287 °C at the hot corner, where the wall temperature is discontinuous (Fig. 15). The stopping error at the center is estimated at about 0.006 °C; this is an estimate, not a measurement. The measured TV decay rates are offset from the slowest-mode decay rate by 0.12 % for FTCS and −0.08 % for ADI. The final probe values sit below the exact ones because the bar is still warming when the run stops.

ADI at $\Delta t = 1.0$ s stopped later, at 496 s versus 411 s. Its probe values ended closer to the exact ones, but its center-node error grew, 0.0046 °C versus 0.0003 °C.

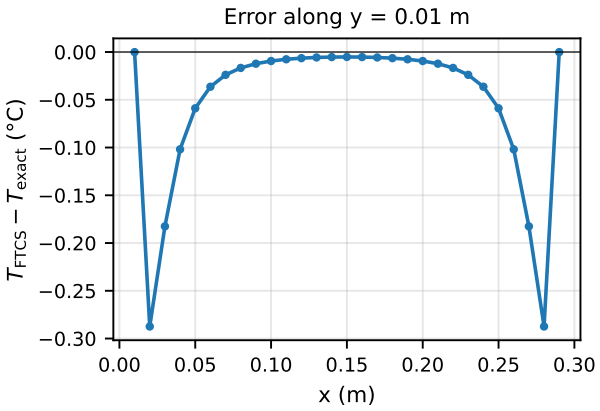



**Fig. 15 FTCS steady-state error along the first interior row, $y = 0.01$ m.**

### C. Efficiency

At $\Delta t = 0.2$ s, FTCS took 2054 steps to reach steady state and ADI took 2057. ADI also runs at $\Delta t = 1.0$ s, reaching steady state in 496 steps which is about 4.1 times fewer. FTCS cannot since $\Delta t_{max} = 0.2225$ s. Each step costs more for ADI since FTCS does one explicit update per node, while ADI does two Thomas sweeps per node, a few times more work per step. Wall-clock time was not measured.

### D. Comparison of FTCS and ADI

At $\Delta t = 0.2$ s, the ADI and FTCS solutions differ by at most 0.096 °C at $t = 10$ s, near the hot wall, and by 0.026 °C at $t = 40$ s. The ADI profiles match the FTCS ones in Fig. 4, and both methods have the same steady-state error. ADI is stable at $\Delta t = 1.0$ s, with a center temperature of 7.6878 °C versus 7.6833 °C exact, while FTCS is not.

## VIII. Conclusion

Both schemes reach the same steady state, with the same error against the analytical solution: an RMS error of 0.026 °C and a maximum of 0.287 °C at the hot corner. FTCS is stable at $\Delta t = 0.2$ s and diverges at $\Delta t = 1.0$ s, as its stability limit of 0.2225 s predicts. ADI converges at both time steps, reaching steady state in 496 steps at $\Delta t = 1.0$ s, although each step costs more than an FTCS step. The steady-state wall heat flux satisfies the overall energy balance within 0.1 %. The fractional step method was not implemented per the project requirements, so the comparison covers FTCS and ADI only.

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educated on the AIAA format, and made cleanup, formatting, and flow edits to the author's wording. The author wrote the concepts and text of this report, verified all results, and is responsible for all of the content provided.

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